R-LearnXR: Reproducible R-Based Virtual Laboratories for Data Science Education

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Prepared for the R Consortium ISC 2026-2 submission. This document follows the required proposal template revised 2026-03-24. The project can be submitted by the Project Lead; external review and pilot recruitment are described as grant-period validation activities. No external endorsement is claimed here.

Executive Summary

R-LearnXR will provide a small, reusable R package and Quarto authoring framework for reproducible data-science lessons. R educators can readily publish code, text, and two-dimensional graphics, but an interactive multivariate lesson still requires authors to coordinate analysis data, browser rendering, learner prompts, accessible alternatives, environment locking, and publishing. Existing examples are often tied to one course or an opaque hosted service and are difficult for another instructor to adapt.

The project will turn a working prototype into community-ready R infrastructure. Authors will use `scaffold_lesson()`, `render_scene()`, and `check_lesson()` to move from an R data frame to a documented Quarto lesson, a portable scene artifact, a semantic table, and release checks. Learners can predict a result, edit and execute R locally through a pinned WebR runtime, inspect the same data in an optional 3D view or table, explain coordinate evidence, and export R or Quarto source. The 3D/browser layer is evidence for the package contract, not the grant's product claim; the optional AI Visual Brief is explicitly outside the funded MVP.

Grant work will stabilize the package API, validate and extend the three-lesson baseline with a community-authored lesson adaptation, expand reproducibility and browser-accessibility checks, conduct a small educator and learner feasibility pilot, and publish contributor training and maintenance documentation. The prototype reduces delivery risk by demonstrating the package-to-lesson workflow now. We request **$10,000 for six months, January through June 2027**. Existing prototype development is in-kind evidence of feasibility and is not included as retroactive cost.

Signatories

Project team

- **Jewoong Moon, Project Lead and primary contractor.** Responsible for learning-experience design, R and Quarto integration, reproducibility workflow, documentation, community coordination, and project reporting. Track record and current implementation are available in the public R-LearnXR repository.

- **Additional core team members: optional and to be added if confirmed.** The proposal is intentionally scoped so the Project Lead can deliver the six-month MVP as the primary contractor. Any confirmed collaborator will receive a named responsibility and be added with their consent.

Contributors

People who contribute lesson examples, accessibility review, code, or proposal feedback will be named here after confirming how they wish to be credited.

Consulted

The grant-period plan will invite at least two R or Quarto educators or maintainers to review the need, scope, API, and reuse plan. Attributable feedback or public evidence will be linked here when permission is granted. No external endorsement is claimed in this draft.

The Problem

R instructors who want learners to reason about multivariate results must assemble several systems: R data preparation, interactive rendering, explanatory prompts, accessible alternatives, reproducible environments, and publishing. These pieces are usually built ad hoc. The result is duplicated effort, fragile course examples, and activities that other instructors cannot readily inspect or reuse.

The problem affects instructors, package authors, workshop organizers, contributors, and learners. Instructors need a maintainable route from an R analysis to an interactive lesson. Learners need guided evidence-based reasoning instead of a visualization that is merely viewed, plus a non-pointer path when a 3D canvas is inaccessible. Maintainers need examples that can be checked automatically when data, packages, browsers, or publishing tools change.

R already offers strong analysis, visualization, Shiny, WebGL, WebAssembly, and publishing capabilities. R-LearnXR does not replace these projects. Its package-level contribution is a focused educational layer connecting R-generated data, an explicit learning workflow, accessible evidence inspection, reproducibility checks, and contribution practices. The browser-first design makes headsets optional and avoids proprietary application servers. The 3D view is a progressive enhancement; the R data contract and semantic table remain useful without it.

Solving this problem will let the R community share virtual laboratories as open educational resources. An instructor will be able to scaffold a documented template, map three variables from an R data frame into an inspectable scene, add prediction and explanation prompts, verify the build, and publish through an ordinary Quarto workflow. Contributors can improve the package, adapt examples to new domains, translate lessons, and reuse the checks in other educational projects.

The Proposal

Overview

R-LearnXR will become a small R package and Quarto lesson framework for turning analysis results into inspectable learning evidence. R remains the source of data and analysis. The package API is the funded core: `scaffold_lesson()` creates the authoring structure, `render_scene()` validates and serializes a data frame, and `check_lesson()` produces advisory or strict release reports. A pinned WebR runtime and browser view demonstrate the learner-facing contract; the semantic table, exported

R/Quarto source, and static artifacts preserve value when WebR or 3D is unavailable. The optional AI Visual Brief is a documented, unfunded adapter and is not a grant deliverable.

The grant will fund the future work needed to move from prototype to community infrastructure: API stabilization, reusable authoring primitives, a community-authored lesson adaptation, stronger checks, external validation, contributor onboarding, and a documented release process.

Minimum viable product

1. A tested R package with stable functions to scaffold lessons, render browser scenes, and check lesson projects, including edge-case validation and machine-readable reports.
2. A Quarto template implementing Orient, Predict, Run R, Explore, Explain, Transfer, and Reproduce.
3. Three openly licensed example lessons using public datasets, with the current set serving as feasibility evidence and a community-authored adaptation planned during the grant.
4. Strict release checks for substantive data licenses, deterministic seeds, environment locks, portable paths, required learning-loop elements, accessible alternatives, and generated-artifact hashes, plus a real-browser smoke check for keyboard behavior and responsive overflow.
5. An authoring guide, accessibility guide, contributor pathway, pilot protocol, and issue templates.
6. Continuous integration for package tests, lesson checks, and Quarto rendering.

Architecture and dependencies

R prepares and validates author-supplied data. The package writes a small JSON artifact and a maintained HTML/JavaScript laboratory. WebR executes learner-edited R, validates a returned `scene` data frame, and synchronizes it with the canvas and semantic table. Quarto embeds or links the laboratory. A checker produces human-readable reports, and continuous integration repeats the checks and renders examples.

Core dependencies are R, Quarto, WebR, standard browser APIs, `testthat`, and openly licensed datasets. The project will pin tested versions and document every external asset. WebR 0.6.0 is currently loaded from its public runtime URL for first-run browser execution; offline vendor/self-host support is not claimed as complete. No account, proprietary service, headset, or application server is required for learners.

Assumptions, risks, and recovery

- **3D is useful only for selected multivariate tasks.** The pilot will test task fit. The framework will retain table-first and two-dimensional alternatives when 3D adds no instructional value.
- **A pinned WebR runtime remains maintainable.** The release will expose loading and error states and evaluate self-hosted runtime assets. Browser execution is an enhancement; author-side rendering and accessible artifacts remain available.
- **Educators can author concise evidence prompts.** Contributor training will include editable prompt patterns and a review rubric.
- **Scope could expand into a general XR platform.** The grant excludes headset-native applications, multiplayer systems, accounts, cloud analytics, offline WebR vendoring, model training, and a general-purpose 3D grammar. AI-generated briefs remain optional, reviewable, and unfunded.
- **Early adoption may be limited.** Success will use documented reuse intent, pilot completion, outside contribution, and maintainable artifacts, not inflated usage counts.

Project Plan

Start-up and technical delivery

The public repository, licenses, contribution guide, code of conduct, prototype lessons, and continuous-integration workflow already exist. During the first month, the team will confirm roles, publish reporting checkpoints, convert remaining prototype decisions into a release schema, and open milestone issues for community review.

Budgeted deliverable package	Target	Grant-period outcome	Budget
1. Package and API hardening	Jan. 31, 2027	Stable API, schema, edge-case tests, and release criteria	$2,800
2. Reusable lesson system	Mar. 15, 2027	Quarto authoring primitives and one community-authored lesson adaptation	$2,400
3. Reproducibility and browser accessibility	Apr. 15, 2027	Strict JSON/Markdown checks, CI evidence, and browser keyboard/overflow validation	$2,200
4. External validation, release, and maintenance	Jun. 30, 2027	Educator/novice feasibility pilot, contributor training, tagged release, and report	$2,600
Total			**$10,000**

The rows identify scope, outcomes, and partial-funding priorities; they do not request four separate payments, and contract terms govern disbursement. Funds support direct labor during the grant period. No indirect costs, travel, lodging, food, publication fees, or personal hardware are requested. Existing prototype code, UI refinement, narration, screenshots, and the three current lessons are in-kind feasibility evidence and are not charged retroactively.

Community availability and dissemination

Code will remain MIT licensed. Original educational content and templates will use CC BY 4.0, and datasets will retain explicit source licenses. Work will be developed in public issues with contribution templates and quarterly progress notes. The team will solicit feedback from R educators, maintainers, and an R user group or comparable community, prepare an R Consortium blog update, and publish an end-of-project report that connects pilot findings to resolved issues and remaining work.

Success

Definition of done

- Package functions and lesson checks pass in continuous integration.

- Three complete lessons render from documented, locked environments; the grant validates reuse and adds a community-authored adaptation.
- Every example supports keyboard operation and a semantic data alternative.
- Authoring, accessibility, contribution, validation, and maintenance documentation is public.
- A tagged release and final project report are archived and linked from the repository.

Measuring success

- Two R educators or maintainers complete a documented API/scope review, with concrete reuse or review outcomes linked publicly or summarized with permission.
- Five novice learners and two instructors are invited to the structured feasibility protocol; the report records invitations, completed sessions, task times, facilitator interventions, and drop-offs rather than inferring effectiveness from a small sample.
- At least four of five novice learners complete the Predict, Run R, Explore, and Explain loop without facilitator intervention, reported as raw counts with limitations; if recruitment is smaller, the target is reported as not estimable rather than silently changed.
- All critical accessibility defects found in the browser smoke test or external review are fixed or explicitly documented before release.
- At least one contributor outside the core team completes an issue, lesson adaptation, or documentation improvement.

Future work

After the grant, contributors can add domain-specific lesson packs, localization, optional WebXR adapters, and integrations with established R visualization packages. These extensions remain outside the funded MVP so the six-month project can deliver a stable reusable foundation.